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[Portfolio Website](#)

PROFILE

AI/ML Engineer with research experience in Deep Learning, Computer Vision, Explainable AI (XAI), Medical Image Analysis, and Edge AI Deployment. Worked under the guidance of Dr. P. Ranga Babu, Associate Professor, IIITDM Kurnool, on AI-driven healthcare systems involving chest X-ray disease classification, ocular disease diagnosis, oral cancer detection, and explainable AI frameworks. Experienced in developing end-to-end machine learning pipelines from data preprocessing and model development to deployment on embedded and edge devices.

EDUCATION

G. Pulla Reddy Engineering College, Kurnool

Oct 2022 – Jun 2026

Bachelor of Technology (B.Tech)

Electronics and Communication Engineering

CGPA: **8.23** / 10

TECHNICAL SKILLS

Programming Languages: Python, C, MATLAB

Machine Learning & AI: TensorFlow, Keras, Scikit-Learn, Deep Learning, CNNs, Vision Transformers (ViT), Transfer Learning, Explainable AI, Medical Image Analysis

Computer Vision: OpenCV, Image Classification, Feature Extraction, Object Detection, Grad-CAM, SHAP, LIME, Attention Mechanisms (CBAM, SE, BAM)

Deployment: TensorFlow Lite, Flask, Raspberry Pi, Edge AI Deployment

Embedded Systems & IoT: STM32, ESP32, Arduino, MQTT, NodeMCU, Embedded Systems

Tools: Git, Linux, CubeMX, MATLAB, Proteus, Sentaurus TCAD

RESEARCH EXPERIENCE

Research Intern – Medical AI & Computer Vision

Jan 2025 – Jul 2025

Indian Institute of Information Technology Design and Manufacturing (IIITDM) Kurnool

Under the Guidance of Dr. P. Ranga Babu, Associate Professor

- Developed a four-class Chest X-Ray Disease Classification framework for COVID-19, Pneumonia, Tuberculosis, and Normal cases using Wavelet Denoising, Xception, and CBAM Attention.
- Achieved **97.57% classification accuracy**, Macro F1-score of **0.9697**, and optimized deployment on Raspberry Pi using TensorFlow Lite.
- Developed an Explainable AI framework for ocular disease classification using EfficientNet-B3 integrated with Grad-CAM, SHAP, and LIME.
- Achieved **91.6% classification accuracy** while generating clinically interpretable explanations for AI-assisted diagnosis.
- Worked extensively on medical image preprocessing, attention mechanisms, model optimization, explainability, and edge deployment.

PUBLICATIONS

IoT-Based Modular P10 LED Notice Board System for Real-Time Industrial Communication using MQTT and Flask

P. Abdul Khayum, Sreeram Jagadeeshwar, Sadhu Akash, P. Santhana Naik, J. Lingavardhan Reddy

International Journal for Multidisciplinary Research (IJFMR)

Volume 8, Issue 2, March–April 2026

- Designed and deployed a modular industrial communication system for QTPL using ESP8266, MQTT, Flask, and P10 LED Displays.
- Achieved message update latency below 250 ms with uninterrupted operation exceeding 72 hours.
- Reduced deployment cost by more than 80% compared to commercial digital signage systems.
- DOI: [10.36948/IJFMR.2026.V08I02.74943](https://doi.org/10.36948/IJFMR.2026.V08I02.74943)

PROJECTS

Oral Cancer Detection using Deep Learning

- Developed oral cancer classification models using VGG19, MobileNetV2, and Vision Transformer architectures.
- Applied transfer learning, hyperparameter optimization, attention mechanisms, and explainable AI techniques.
- Evaluated performance using ROC-AUC, F1-score, confusion matrix, and Grad-CAM visualizations.

PALS-QTPL Real-Time Digital Notice Board Using MQTT

- Secured **Rs.80,000 innovation funding jointly supported by PALS (Pan IIT Alumni Leadership Series), IIT Madras (IITM), and IIT Hyderabad (IITH)** for the design and deployment of an industrial IoT communication system.
- Led development of an ESP8266-based industrial communication platform using MQTT and Flask.
- Successfully deployed the system in an industrial environment and validated performance requirements.
- Resulted in a peer-reviewed journal publication.

Gesture-Controlled Virtual Mouse Using Computer Vision

- Developed a touchless human-computer interaction system using OpenCV and hand gesture recognition.
- Enabled cursor movement and click operations through real-time image processing.

RESEARCH PROJECTS

Wavelet-Denoised Chest X-Ray Disease Classification Using Xception with CBAM Attention

- Four-class disease classification framework for COVID-19, Pneumonia, Tuberculosis, and Normal chest X-rays.
- Achieved 97.57% accuracy and successful TensorFlow Lite deployment on Raspberry Pi.
- Manuscript under preparation for journal/conference submission.

Explainable Deep Learning for Multi-Class Ocular Disease Classification from Fundus Images

- Developed an EfficientNet-B3 based explainable AI framework.
- Integrated Grad-CAM, SHAP, and LIME for interpretable medical diagnosis.
- Achieved 91.6% classification accuracy and Macro F1-score of 0.925.
- Manuscript under preparation for journal/conference submission.

Oral Cancer Detection Using Deep Learning and Vision Transformers

- Investigated CNN and Vision Transformer architectures for oral cancer classification.
- Evaluated transfer learning, attention mechanisms, explainability, and lightweight deployment strategies.
- Research work currently being prepared for publication.

CERTIFICATIONS

- Microchip Embedded System Developer Certification
- Sentaurus TCAD Training (ISWDP, LV1, LV2, Advanced)
- NPTEL – Introduction to Internet of Things

LEADERSHIP & ACHIEVEMENTS

- IEEE MTT-S Society Secretary.
- Secured **Rs.80,000 funding** through the PALS-QTPL Industry Innovation Program.
- Selected for IEEE YESIST International Project Showcase.
- First Prize – State Level Anokha Idea Presentation.
- Second Prize – Recent Technologies: IoT and Its Uses.
- Top Performer – NPTEL Introduction to IoT.